作业

作业来自

梁昆淼《力学》 第 4 版 高等教育出版社

Johns Analytical Mechanics 2nd ed Oxford University Press

例如： L01.08 表示梁昆淼书第 01 章第 08 题； J02.05 Johns 书第 02 章第 05 题

J03.07† 表示 Johns 书第 03 章第 07 题，带 † 表示仅要求荣誉班同学完成

Homework 03, Week 04 2023.09.25 Due 2023.10.09 J02.05, J02.06, J02.09, J02.12

★ J02.05

Suppose that two masses have a motion defined by a Lagrangian function

$$L = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 - U(|\vec{r}_1 - \vec{r}_2|)$$

where $\vec{r}_1$ and $\vec{r}_2$ denote the position vectors of masses 1 and 2, respectivley,

$$\vec{v}_1 = \frac{d\vec{r}_1}{dt}, \quad \vec{v}_2 = \frac{d\vec{r}_2}{dt}, \quad v_1^2 = \vec{v}_1 \cdot \vec{v}_1, \quad v_2^2 = \vec{v}_2 \cdot \vec{v}_2.$$

Let $\vec{r} = \vec{r}_2 - \vec{r}_1, \quad \vec{R} = \dfrac{m_2 \vec{r}_2 + m_1 \vec{r}_1}{m_1 + m_2}$

(*a*) Rewrite the Lagrangian in terms of the variables $\vec{r}$ and $\vec{R}$ and their time derivatives.

Show the Lagrangian can be written as two terms, one of which depends only on $\vec{R}$ and its time derivative and the other only on $\vec{r}$ and its time derivative. Such Lagrangian system are called separable.

(*b*) Show that three components of $\vec{R}$ are ignorable coordinates and that the total momentum of the system is conserved.

Solution:

(*a*)

$$L = T - U = \frac{1}{2} m_1 \vec{v}_1^2 + \frac{1}{2} m_2 \vec{v}_2^2 - U(|\vec{r}_1 - \vec{r}_2|)$$

$$\begin{cases} \vec{r} = \vec{r}_2 - \vec{r}_1 \\ \vec{R} = \dfrac{m_2 \vec{r}_2 + m_1 \vec{r}_1}{m_1 + m_2} \end{cases} \Longrightarrow \begin{cases} \vec{r}_1 = \vec{R} - \dfrac{m_2}{M} \vec{r} \\ \vec{r}_2 = \vec{R} + \dfrac{m_1}{M} \vec{r} \end{cases} \Longrightarrow \begin{cases} \vec{v}_1 = \vec{V} - \dfrac{m_2}{M} \vec{v} \\ \vec{v}_2 = \vec{V} + \dfrac{m_1}{M} \vec{v} \end{cases}$$

where $M = m_1 + m_2, \quad \vec{v} = \dot{\vec{r}}$

So $\quad m_1 \vec{v}_1^2 + m_2 \vec{v}_2^2 = M \vec{V}^2 + \eta \vec{v}^2, \quad$ with $\eta = \dfrac{m_1 m_2}{M}$

$$\Longrightarrow \quad L = \frac{1}{2} M \vec{V}^2 + \frac{1}{2} \eta \vec{v}^2 - U(r)$$

where the first term depends only on $\vec{V}$, and the seconde term on $\vec{r}$ and $\vec{v}$

(*b*)

L is indepent of $\vec{R}$, so $\vec{R}$ are ignorable coordinates

$$\frac{d}{dt}\frac{\partial L}{\partial \vec{V}} - \frac{\partial L}{\partial \vec{R}} = 0 \quad \Longrightarrow \quad \frac{d}{dt}\frac{\partial L}{\partial \vec{V}} = 0$$

But $\dfrac{\partial L}{\partial \vec{V}} = M\,\vec{V}$ $\Longrightarrow$ The total momentum $M\,\vec{V} = \vec{P} = m_1\,\vec{v}_1 + m_2\,\vec{v}_2$ is conserved

★ J02.06

A mass m is acted on by a force derived from the generalized potential

$U^{(\text{vel})}(\vec{r},\,\vec{v},\,t) = V(r) + \vec{\sigma}\cdot\vec{L}$ with the superscript "(vel)" denoting velocity dependent potential

where $r = \sqrt{x^2 + y^2 + z^2}$, $\vec{\sigma} = \sigma\,\vec{e}_z$, $\vec{L} = \vec{r}\times m\,\vec{v}$ and $\vec{r}$ and $\vec{v}$ are the position and velocity of the mass relative to some inertial coordinate system.

(a) Express $U^{(\text{vel})}$ in Cartesian coordinates (the s system) $x,\ y,\ z,\ \dot{x},\ \dot{y},\ \dot{z}$ and find the force $\vec{F}$ and its three Cartesian components $F_x,\ F_y,\ F_z$

(b) Express $U^{(\text{vel})}$ in spherical polar coordinates (that can be called the q system) $r,\ \theta,\ \phi,\ \dot{r},\ \dot{\theta},\ \dot{\phi}$ and find the generalized force $Q_k,\ k = 1,\ 2,\ 3$

(c) The force vector found in (a) can be rewritten in terms of spherical polar unit vectors as

$\vec{F} = F_r\,\vec{e}_r + F_\theta\,\vec{e}_\theta + F_\phi\,\vec{e}_\phi$ with $F_r = \vec{e}_r\cdot\vec{F}$, $F_\theta = \vec{e}_\theta\cdot\vec{F}$, $F_\phi = \vec{e}_\phi\cdot\vec{F}$. Show $Q_r = F_r$.

(d) However $Q_\phi \neq F_\phi$, what does Q_ϕ correspond to in Newton Mechanics? What is the dimensionality of $Q_\phi\,\delta\phi$?

(e) Based on this example, what do you learn about the difference between the orthogonal curvilinear coordinate system and the generalized coordinates?

Solution:

(a)

$$U^{(\text{vel})}(\vec{r},\,\vec{v},\,t) = V(r) + \vec{\sigma}\cdot\vec{L}$$

$$\begin{aligned} U(x,\,y,\,z,\,\dot{x},\,\dot{y},\,\dot{z},\,t) &= V\left[(x^2+y^2+z^2)^{1/2}\right] + m\,\sigma\,\vec{e}_z\cdot(\vec{r}\times\vec{v}) \\ &= V\left[(x^2+y^2+z^2)^{1/2}\right] + m\,\sigma(x\,\dot{y} - y\,\dot{x}) \end{aligned}$$

Cartesian components $F_\alpha = \dfrac{d}{dt}\left(\dfrac{\partial U}{\partial \dot{s}_\alpha}\right) - \dfrac{\partial U}{\partial s_\alpha}$

For $\alpha = 1$: $\left(\dfrac{\partial U}{\partial \dot{x}}\right) = -m\,\sigma\,y \quad \Longrightarrow \quad \dfrac{d}{dt}\left(\dfrac{\partial U}{\partial \dot{x}}\right) = -m\,\sigma\,\dot{y}$

$$\left(\frac{\partial U}{\partial x}\right) = m\,\sigma\,\dot{y} + \frac{\partial V}{\partial x} = m\,\sigma\,\dot{y} + V'(r)\,\frac{x}{r}$$

$$\Longrightarrow \quad F_x = -2\,m\,\sigma\,\dot{y} - \frac{x}{r}\,V'(r)$$

For $\alpha = 2$: $F_y = 2\,m\,\sigma\,\dot{x} - \dfrac{y}{r}\,V'(r)$

For $\alpha = 3$: $F_z = -\dfrac{z}{r}\,V'(r)$

(b)

In the spherical coordinates, one has

$$\begin{cases} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{cases} \quad \Longrightarrow \quad \begin{cases} \dot{x} = \dot{r}\sin\theta\cos\phi + r\,\dot{\theta}\cos\theta\cos\phi - r\,\dot{\phi}\sin\theta\sin\phi \\ \dot{y} = \dot{r}\sin\theta\sin\phi + r\,\dot{\theta}\cos\theta\sin\phi + r\,\dot{\phi}\sin\theta\cos\phi \end{cases}$$

$$U(x, y, z, \dot{x}, \dot{y}, \dot{z}, t) = V\left[\left(x^2 + y^2 + z^2\right)^{1/2}\right] + m\,\sigma\left(x\,\dot{y} - y\,\dot{x}\right)$$
$$= V(r) + m\,\sigma\,r^2\,\dot{\phi}\sin^2\theta$$

Below are *Mathematica* codes for evaluating $\vec{e}_z\cdot\left(\vec{r}\times\vec{v}\right)$

```
x = r[t] Sin[θ[t]] Cos[ϕ[t]];
y = r[t] Sin[θ[t]] Sin[ϕ[t]];
z = r[t] Cos[θ[t]];
xdot = D[x, t];
ydot = D[y, t];
zdot = D[z, t];

res = {0, 0, 1}.({x, y, z} × {xdot, ydot, zdot});
(* use "ESC cross ESC" to input symbol of cross product *)

Simplify[res]
```

```
r[t]^2 Sin[θ[t]]^2 ϕ′[t]
```

Alternatively, , you can derive $\vec{e}_z\cdot\left(\vec{r}\times\vec{v}\right)$ as follows

$$\vec{v} = \dot{r}\,\vec{e}_r + r\,\dot{\theta}\,\vec{e}_\theta + r\,\dot{\phi}\sin\theta\,\vec{e}_\phi,\quad \vec{r} = r\,\vec{e}_r,\quad \vec{e}_z = \cos\theta\,\vec{e}_r - \sin\theta\,\vec{e}_\theta$$

Triple scalar product is then $\vec{e}_z\cdot\left(\vec{r}\times\vec{v}\right) = \begin{vmatrix} \cos\theta & -\sin\theta & 0 \\ r & 0 & 0 \\ \dot{r} & r\,\dot{\theta} & r\,\dot{\phi}\sin\theta \end{vmatrix} = \left(r^2\sin^2\theta\right)\dot{\phi}$

Generalized force : remember we have $U = V(r) + m\,\sigma\,r^2\,\dot{\phi}\sin^2\theta$

$$Q_\alpha = \frac{d}{dt}\left(\frac{\partial U}{\partial \dot{q}_\alpha}\right) - \frac{\partial U}{\partial q_\alpha} \implies \begin{cases} Q_r = -2\,m\,\sigma\,r\sin^2\theta\;\dot{\phi} - V'(r) \\ Q_\theta = -2\,m\,\sigma\,r^2\sin\theta\cos\theta\,\dot{\phi} \\ Q_\phi = \dfrac{d}{dt}\left(m\,\sigma\,r^2\sin^2\theta\right) \end{cases}$$

(*c*)

$$\vec{F} = F_x\,\vec{e}_x + F_y\,\vec{e}_y + F_z\,\vec{e}_z,\ \text{with}\ \begin{cases} F_x = -2\,m\,\sigma\,\dot{y} - \dfrac{x}{r}\,V'(r) \\ F_y = 2\,m\,\sigma\,\dot{x} - \dfrac{y}{r}\,V'(r) \\ F_z = -\dfrac{z}{r}\,V'(r) \end{cases}$$

$$F_r = \vec{e}_r\cdot\vec{F} = F_x\sin\theta\cos\phi + F_y\sin\theta\sin\phi + F_z\cos\theta$$

Let us simplify it directly by *Mathematica*

```
Fx = -2 m σ ydot - x/r Vp;  (* for simplicity, we use Vp to denote V′(r)  *)

Fy = 2 m σ xdot - y/r Vp;  (* here xdot and ydot are ẋ and ẏ, given above *)

Fz = -z/r Vp;

sim = {r[t] → r, θ[t] → θ, ϕ[t] → ϕ, ϕ′[t] → ϕdot};

Fr = Fx Sin[θ] Cos[ϕ] + Fy Sin[θ] Sin[ϕ] + Fz Cos[θ];
Fr = Simplify[Fr /. sim]

Qr = -2 m σ r Sin[θ]^2 ϕdot - Vp;

Simplify[Fr - Qr]
```

```
-Vp - m r σ ϕdot + m r σ ϕdot Cos[2 θ]
```

```
0
```

(*d*)

Torque $\vec{\tau} = \vec{r} \times \vec{F}$, z-component $\tau_z = \vec{e}_z \cdot \left(\vec{r} \times \vec{F}\right) = \begin{vmatrix} 0 & 0 & 1 \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix} = x F_y - y F_x$

$$\tau_z = 2 m \sigma x \dot{x} - \frac{x y}{r} V'(r) - \left[-2 m \sigma y \dot{y} - \frac{x y}{r} V'(r)\right] = 2 m \sigma \left(x \dot{x} + y \dot{y}\right) = m \sigma \frac{d}{dt}\left(x^2 + y^2\right)$$

$$x^2 + y^2 = r^2 \sin^2 \theta \quad \Longrightarrow \quad \tau_z = m \sigma \frac{d}{dt}\left(r^2 \sin^2 \theta\right) = Q_\phi$$

$$[Q_\phi] = \frac{m\,[\sigma]\,L^2}{T}, \quad [q_\phi] = [\phi] = 1 \quad \Longrightarrow \quad [Q_\phi][\delta\phi] = \frac{m\,[\sigma]\,L^2}{T}$$

From the expression of generalized potential, we know

$[U] = \dfrac{m\,[\sigma]\,L^2}{T}$ so $[Q_\phi][\delta\phi] = \dfrac{m\,[\sigma]\,L^2}{T}$ has the dimentionality of energy

Confirmed $\tau_z = Q_\phi$ by *Mathematica*

```
ez = {0, 0, 1};
rv = {x, y, z};
Fv = {Fx, Fy, Fz};
tz = Dot[ez, Cross[rv, Fv]];  (* Another way to input scalar triple product *)
tz = FullSimplify[tz /. sim]

Qϕ = D[m σ r[t]^2 Sin[θ[t]]^2, t];
Simplify[(tz - Qϕ) /. sim]
```

```
2 m r σ Sin[θ] (Sin[θ] r′[t] + r Cos[θ] θ′[t])
```

```
0
```

(e)

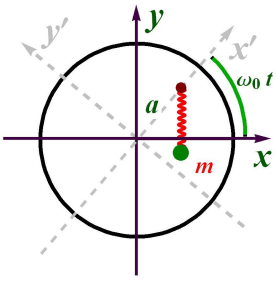
For example

(1) 正交曲线坐标系中，力还是力，具有力的量纲，在广义坐标中，广义力未必有力的量纲

(2) 在广义坐标中，广义力与广义坐标之乘积的量纲总是与功（能量）的量纲相同

★ J02.09

A horizontal, circular table with a frictionless top surface is constrained to rotate about a vertical line through its center, with constant angular velocity ω_0. A peg is driven into the table top at a distance a from the center of the circle. A mass m slides freely on the top surface of the table, connected to the peg by a massless spring of force constance k and zero rest length. Take the s system to be an inertial system of Cartesian coordinates x, y with the origin at the center of the table top, and the q system to be rotating Carteisan coordinates x', y' defined so that $\vec{e}'_x$ defines a line passing through the peg. Ignore the z coordinate, and treat this problem as one with two degrees of freedom.The transformation between coordinates of the mass in the two systems is $x = x' \cos\omega_0 t - y' \sin\omega_0 t$, $y = x' \sin\omega_0 t + y' \cos\omega_0 t$



(a) Wirte $L(s, \dot{s}, t)$ in the s system and $L(q, \dot{q}, t)$ in the q system.

(b) Write H_s in the s system. Is it equal to $T + U$? Is it conserved?

(c) Write H_q in the q system. Is it equal to $T + U$? Is it conserved?

(d) What do you learn about the conserveness and generalized energy function?

Solution:

Let $s_1 = x$, $s_2 = y$ and $q_1 = x'$, $q_2 = y'$

$$\begin{cases} x = x' \cos\omega t - y' \sin\omega t \\ y = x' \sin\omega t + y' \cos\omega t \end{cases} \quad (1) \quad \Longrightarrow \quad \begin{cases} x' = x \cos\omega t + y \sin\omega t \\ y' = -x \sin\omega t + y \cos\omega t \end{cases} \quad (2)$$

(a)

In s-system : $T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2)$, $\qquad V(s, t) = ?$

In q-system : $V(q, t) = \frac{1}{2} k \left[(x' - a)^2 + y'^2\right]$, $\quad T(q, \dot{q}\, t) = ?$

We know $V(q, t)$ is a scalar, so $V(s, t) = V(q(s, t), t)$

Then in s-system $V(s, t) = \frac{k}{2}\left[(x \cos\omega t + y \sin\omega t - a)^2 + (-x \sin\omega t + y \cos\omega t)^2\right]$

$$L(s, \dot{s}, t) = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - \frac{k}{2}\left[(x \cos\omega t + y \sin\omega t - a)^2 + (-x \sin\omega t + y \cos\omega t)^2\right]$$

The Lagrangian equations yield

$$\begin{cases} m\ddot{x} + k x = k a \cos\omega t \\ m\ddot{y} + k y = k a \sin\omega t \end{cases} \Longrightarrow \begin{cases} x = \dfrac{\omega_0^2 a}{(\omega_0^2 - \omega^2)} \cos\omega t - \dfrac{\omega^2 a}{(\omega_0^2 - \omega^2)} \cos\omega_0 t \\ y = \dfrac{\omega_0^2 a}{(\omega_0^2 - \omega^2)} \sin\omega t - \dfrac{\omega_0 \omega a}{(\omega_0^2 - \omega^2)} \sin\omega_0 t \end{cases}$$

where the initial conditions are $\begin{cases} x(0) = a \\ \dot{x}(0) = 0 \end{cases}$, $\begin{cases} y(0) = 0 \\ \dot{y}(0) = 0 \end{cases}$ and $\omega_0^2 = \dfrac{k}{m}$

```
Quit
```

```
diffx = m x''[t] + k x[t] - k a Cos[w t];
diffy = m y''[t] + k y[t] - k a Sin[w t];

sol = DSolve[diffx == 0, x[t], t];
resx = Simplify[x[t] /. sol[[1]]];

eq1 = (resx /. t → 0) - a;
eq2 = (D[resx, t] /. t → 0);

solcx = Solve[{eq1 == 0, eq2 == 0}, {C[1], C[2]}];
xs = resx /. solcx[[1]] /. {√k → w0 √m, k → w0^2 m};
xs = Simplify[xs]


sol = DSolve[diffy == 0, y[t], t];
resy = Simplify[y[t] /. sol[[1]]];

eq1 = (resy /. t → 0);
eq2 = (D[resy, t] /. t → 0);

solcy = Solve[{eq1 == 0, eq2 == 0}, {C[1], C[2]}];
ys = resy /. solcy[[1]] /. {√k → w0 √m, k → w0^2 m};
ys = Simplify[ys]

{Simplify[xs /. t → 0], Simplify[ys /. t → 0],
 Simplify[D[xs, t] /. t → 0], Simplify[D[ys, t] /. t → 0]}
```

```
(a (-w0^2 Cos[t w] + w^2 Cos[t w0])) / (w^2 - w0^2)
```

```
(a w0 (w0 Sin[t w] - w Sin[t w0])) / (-w^2 + w0^2)
```

```
{a, 0, 0, 0}
```

Now we get $L(q, t)$ in q-system, which can be simply done by using Equation (1) and their time derivatives

$$\begin{cases} \dot{x} = \dot{x}' \cos\omega t - \dot{y}' \sin\omega t - \omega x' \sin\omega t - \omega y' \cos\omega t \\ \dot{y} = \dot{x}' \sin\omega t + \dot{y}' \cos\omega t + \omega x' \cos\omega t - \omega y' \sin\omega t \end{cases}$$

$$T(q, \dot{q}, t) = \frac{m}{2}\left[(\dot{x}'^2 + \dot{y}'^2) + \omega^2(x'^2 + y'^2) + 2\omega(x'\dot{y}' - y'\dot{x}')\right]$$

$$L(q, \dot{q}, t) = \frac{m}{2}\left[(\dot{x}'^2 + \dot{y}'^2) + \omega^2(x'^2 + y'^2) + 2\omega(x'\dot{y}' - y'\dot{x}')\right] - \frac{1}{2}k\left[(x' - a)^2 + y'^2\right],$$

The Lagrangian equations yield

$$\begin{cases} m\,\ddot{x}' - m\,\omega^2\,x' - 2\,m\,\omega\,\dot{y}' = -k\,(x' - a) \\ m\,\ddot{y}' - m\,\omega^2\,y' + 2\,m\,\omega\,\dot{x}' = -k\;y' \end{cases}$$

$$\Longrightarrow \begin{cases} x' = \dfrac{\omega_0^2\,a}{\omega_0^2 - \omega^2} - \dfrac{a\,\omega^2}{\omega_0^2 - \omega^2}\cos\omega\,t\cos\omega_0\,t - \dfrac{a\,\omega\,\omega_0}{\omega_0^2 - \omega^2}\sin\omega\,t\sin\omega_0\,t \\ y' = -\dfrac{a\,\omega\,\omega_0}{\omega_0^2 - \omega^2}\cos\omega\,t\sin\omega_0\,t + \dfrac{a\,\omega^2}{\omega_0^2 - \omega^2}\sin\omega\,t\cos\omega_0\,t \end{cases}$$

where the initial conditions are $\begin{cases} x'(0) = a \\ \dot{x}'(0) = 0 \end{cases}$, $\begin{cases} y'(0) = 0 \\ \dot{y}'(0) = -\omega\,a \end{cases}$ and $\omega_0^2 = \dfrac{k}{m}$

```
diffeqx = m x''[t] - m w^2 x[t] - 2 m w y'[t] + k (x[t] - a);
diffeqy = m y''[t] - m w^2 y[t] + 2 m w x'[t] + k y[t];
diffeq = m z''[t] - m w^2 z[t] + 2 m w ⅈ z'[t] + k (z[t] - a);
sol = DSolve[diffeq == 0, z[t], t];    sol = Simplify[sol, {m > 0, a > 0, k > 0, w > 0, t > 0}];
res = (z[t] /. sol[[1]]);
eq1 = (res /. t → 0) - a;   eq2 = (D[res, t] /. t → 0) + ⅈ w a;
solc = Solve[{eq1 == 0, eq2 == 0}, {C[1], C[2]}];   solc = Simplify[solc];

tst = diffeq /. {z[t] → res, z'[t] → D[res, t], z''[t] → D[res, {t, 2}]};   Simplify[tst];

xp = Expand[Simplify[ComplexExpand[Re[res /. solc[[1]]]], {k > 0, m > 0, a > 0}]]
yp = Expand[Simplify[ComplexExpand[Im[res /. solc[[1]]]], {k > 0, m > 0, a > 0}]]
{Simplify[xp /. t → 0], Simplify[yp /. t → 0],
 Simplify[D[xp, t] /. t → 0], Simplify[D[yp, t] /. t → 0, {k > 0, m > 0, a > 0}]}
simp = {x[t] → xp, x'[t] → D[xp, t], x''[t] → D[xp, {t, 2}],
   y[t] → yp, y'[t] → D[yp, t], y''[t] → D[yp, {t, 2}]};
tstx = diffeqx /. simp;   tsty = diffeqy /. simp;   Simplify[{tstx, tsty}, {k > 0, m > 0, a > 0}]
```

$$\frac{a\,k}{k - m\,w^2} - \left(a\,m\,w^2\,\mathrm{Cos}\left[\sqrt{\frac{k}{m}}\;t\right]\mathrm{Cos}[t\,w]\right)\Big/\left(k - m\,w^2\right) - \left(a\,\sqrt{k\,m}\;w\,\mathrm{Sin}\left[\sqrt{\frac{k}{m}}\;t\right]\mathrm{Sin}[t\,w]\right)\Big/\left(k - m\,w^2\right)$$

$$\left(a\,\sqrt{\frac{k}{m}}\;m\,w\,\mathrm{Cos}[t\,w]\,\mathrm{Sin}\left[\sqrt{\frac{k}{m}}\;t\right]\right)\Big/\left(-k + m\,w^2\right) - \left(a\,m\,w^2\,\mathrm{Cos}\left[\sqrt{\frac{k}{m}}\;t\right]\mathrm{Sin}[t\,w]\right)\Big/\left(-k + m\,w^2\right)$$

```
{a, 0, 0, -a w}
```

```
{0, 0}
```

```
zs = xp + ⅈ yp;
tst = diffeq /. {z[t] → zs, z'[t] → D[zs, t], z''[t] → D[zs, {t, 2}]};
tz = Simplify[tst, {k > 0, m > 0, a > 0}];
tst = diffx /. {x[t] → xs, x''[t] → D[xs, {t, 2}]};   tx = Simplify[tst];
tst = diffy /. {y[t] → ys, y''[t] → D[ys, {t, 2}]};   ty = Simplify[tst];
{tx, ty, tz}
```

```
{0, 0, 0}
```

$$\begin{cases} x' = x\cos\omega t + y\sin\omega t \\ y' = -x\sin\omega t + y\cos\omega t \end{cases}$$

```
eqx = xp - xs Cos[w t] - ys Sin[w t];
eqy = yp + xs Sin[w t] - ys Cos[w t];
{FullSimplify[eqx, {k > 0, m > 0}], FullSimplify[eqy, {k > 0, m > 0}]}
```

```
{0, 0}
```

(*b*)

$$\begin{aligned} H_s &= \sum_\alpha \frac{\partial L}{\partial \dot{s}_\alpha}\dot{s}_\alpha - L(s, \dot{s}, t) \\ &= \frac{1}{2} m\left(\dot{x}^2 + \dot{y}^2\right) + \frac{k}{2}\left[(x\cos\omega t + y\sin\omega t - a)^2 + (-x\sin\omega t + y\cos\omega t)^2\right] \\ &= T + U \end{aligned}$$

H_s is not conserved, as can be seen from

$$\dot{H}_s = -\frac{\partial L(s, \dot{s}, t)}{\partial t} \neq 0$$

(*c*)

$$\begin{aligned} H_q &= \sum_\alpha \frac{\partial L}{\partial \dot{q}_\alpha}\dot{q}_\alpha - L(q, \dot{q}, t) \\ &= \frac{1}{2} m\left(\dot{x}'^2 + \dot{y}'^2\right) \underbrace{- \frac{1}{2} m\,\omega^2\left(x'^2 + y'^2\right)}_{T_0} + \frac{k}{2}\left[(x' - a)^2 + y'^2\right] \\ &= T + U - T_0 \neq T + U \end{aligned}$$

However, $\dot{H}_q = -\dfrac{\partial L(q, \dot{q}, t)}{\partial t} = 0$ so H_q is conserved.

That means $H(q, \dot{q}, t) = H(q(0), \dot{q}(0), 0)$, or

$$H(q, \dot{q}, t) = \frac{1}{2} m\left(\dot{x}_0'^2 + \dot{y}_0'^2\right) - \frac{1}{2} m\,\omega^2\left(x_0'^2 + y_0'^2\right) + \frac{k}{2}\left[(x_0' - a)^2 + y_0'^2\right]$$

(*d*) For example

	$\dfrac{\partial q(s, t)}{\partial t} = 0$ $H_q = T + U$	$\dfrac{\partial q(s, t)}{\partial t} \neq 0$ $H_q \neq T + U$
$\dfrac{\partial L}{\partial t} = 0$ H conserved		H_q
$\dfrac{\partial L}{\partial t} \neq 0$ H not conserved	H_s	

已求出，在 s-系 $(s_1 = x,\ s_2 = y)$，拉格朗日量为 $(s_1 = x,\ s_2 = y)$

$$L(s, \dot{s}, t) = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - \frac{k}{2}[(x \cos\omega t + y \sin\omega t - a)^2 + (-x \sin\omega t + y \cos\omega t)^2] \qquad (\dagger)$$

为了求 q-系 $(q_1 = x',\ q_2 = y')$ 的拉格朗日量，我们将 $s = s(q, t),\ \dot{s} = \dot{s}(q, \dot{q}, t)$，即

$$\begin{cases} x = x' \cos\omega t - y' \sin\omega t \\ y = x' \sin\omega t + y' \cos\omega t \end{cases} \text{ and thus } \begin{cases} \dot{x} = \dot{x}' \cos\omega t - \dot{y}' \sin\omega t - \omega x' \sin\omega t - \omega y' \cos\omega t \\ \dot{y} = \dot{x}' \sin\omega t + \dot{y}' \cos\omega t + \omega x' \cos\omega t - \omega y' \sin\omega t \end{cases}$$

代入 $L(s, \dot{s}, t)$ 得

$$L(q, \dot{q}, t) = \frac{m}{2}[(\dot{x}'^2 + \dot{y}'^2) + \omega^2(x'^2 + y'^2) + 2\omega(x'\dot{y}' - y'\dot{x}')] - \frac{1}{2} k [(x' - a)^2 + y'^2] \qquad (\ddagger)$$

上式即为 q-系的拉格朗日量。它是经过坐标变换得到的。

从以上计算中我们得知，q-系只是一种坐标的代换，并没有参考系的变换。

我们依然在实验室参考系中观察，只不过使用了 q 坐标，而 s 坐标与 q 坐标的变换显含了时间：$s = s(q, t)$

若要直接在 q 系中写出拉格朗日量，并不是要换一个参考系，在新的参考系中写出拉格朗日量。

因为新旧参考系观察，速度是不同的，不能保证速度相同（势能也未必相等），故拉格朗日量未必能相等。

而 s-系和 q-系，可以视为同一个实验室参考系观察，使用了不同的坐标（例如直角坐标和球坐标）。

因此，若想直接在 q-系中写出拉格朗日量，那么，速度必须仍然是在实验室参考系观察的速度。

注意参考系、坐标系两个概念不尽相同，尽管许多情况下可以混用。

而拉格朗日量依然是动能 T 减势能 U。只不过坐标不再使用 s-坐标，而是直接使用 q-坐标。

在刚体一章，将会学到，在实验室参考系观察的速度 $\vec{v}$，与在旋转系中观察的速度 $\vec{v}'$，有以下关系

$$\vec{v} = \vec{v}' + \vec{\omega} \times \vec{r}', \quad \text{其中} \quad \vec{v}' = \dot{x}'\vec{e}'_x + \dot{y}'\vec{e}'_y, \quad \vec{r}' = x'\vec{e}'_x + y'\vec{e}'_y, \quad \vec{\omega} = \omega \vec{e}'_z$$

上式的意义是：在实验室参考系观察的速度 $\vec{v} = \dot{x}\vec{e}_x + \dot{y}\vec{e}_y$

等于在旋转参考系观察的速度 $\vec{v}'$，加上因旋转导致的牵连速度 $\vec{\omega} \times \vec{r}'$

从而，用 q-坐标描述，动能为

$$\begin{aligned} T &= \frac{m}{2}\vec{v}^2 = \frac{m}{2}(\vec{v}' + \vec{\omega} \times \vec{r}')^2 = \frac{m}{2}[\dot{x}'\vec{e}'_x + \dot{y}'\vec{e}'_y + \omega\vec{e}'_z \times (x'\vec{e}'_x + y'\vec{e}'_y)]^2 \\ &= \frac{m}{2}[(\dot{x}' - \omega y')\vec{e}'_x + (\dot{y}' + \omega x')\vec{e}'_y]^2 = \frac{m}{2}(\dot{x}' - \omega y')^2 + \frac{m}{2}(\dot{y}' + \omega x')^2 \\ &= \frac{m}{2}[(\dot{x}'^2 + \dot{y}'^2) + \omega^2(x'^2 + y'^2) + 2\omega(x'\dot{y}' - y'\dot{x}')] \end{aligned}$$

势能仍然决定于弹簧的拉长量，为：$V = \frac{1}{2} k [(x' - a)^2 + y'^2]$

从而 q-系的拉格朗日量 $L = T - V$ 为：

$$L(q, \dot{q}, t) = T - V = \frac{m}{2}[(\dot{x}'^2 + \dot{y}'^2) + \omega^2(x'^2 + y'^2) + 2\omega(x'\dot{y}' - y'\dot{x}')] - \frac{1}{2} k [(x' - a)^2 + y'^2] \qquad (\#)$$

上式 (#) 与从 s-系到 q-系经坐标变换得到的拉格朗日量 (‡) 式完全相同。

当然，从 q-系的拉格朗日方程得到的 q-坐标运动方程，

与在旋转坐标系（非惯性系）得到的包含惯性离心力、科里奥利力的“牛二”方程完全相同。

你是否 appreciate 拉格朗日理论的“螓首蛾眉，巧笑倩兮，美目盼兮。”

★ **J02.12**

A point particle of mass m and charge q moves near a very long wire carrying a current I.

Choose the $\vec{e}_z$ axis along the wire in the direction of the current. In the region near the wire, the vector potential in terms of cylindrical polar coordinates ρ, ϕ, z is $\vec{A} = \frac{I}{2\pi c}\ln\left(\frac{\rho}{\rho_0}\right)\vec{e}_z$, where ρ_0 is some arbitrarily chosen ρ value. Assume the electric potential Φ to be zero.

(*a*) Write the Lagrangian $L = L(\rho, \phi, z, \dot{\rho}, \dot{\phi}, \dot{z}, t)$ for the particle, using cylindrical coordinates

(*b*) ind the generalized momenta p_ρ, p_ϕ, p_z

(*c*) Write three Lagrange equations, and show that ϕ and z are ignorable coordinates

(*d*) Use the ϕ and z Lagrange equations to write expressions for $\dot{\phi}$ and $\dot{z}$ as a function of ρ and integration constants

(*e*) Write the generalized energy function. Is it conserved? Use it to express $\dot{\rho}^2$ as a function of ρ and some constants that can be determined at $t = 0$.

Solution:

(*a*)

$$U = q\Phi - q\vec{v}\cdot\vec{A} \quad \text{since } \Phi = 0, \text{ one has } U = -q\vec{v}\cdot\vec{A} \text{ with } \vec{A} = \frac{I}{2\pi c}\ln\left(\frac{\rho}{\rho_0}\right)\vec{e}_z$$

In cylindrical coordinate system $\vec{v} = \dot{\rho}\,\vec{e}_\rho + \rho\,\dot{\phi}\,\vec{e}_\phi + \dot{z}\,\vec{e}_z$

$$L = T - U = \frac{1}{2}m\left(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + \dot{z}^2\right) + \frac{I q \dot{z}}{2\pi c}\ln\left(\frac{\rho_0}{\rho}\right)$$

(*b*)

$$p_\rho = \frac{\partial L}{\partial \dot{\rho}} = m\dot{\rho}, \quad p_\phi = \frac{\partial L}{\partial \dot{\phi}} = m\rho^2\dot{\phi}, \quad p_z = \frac{\partial L}{\partial \dot{z}} = m\dot{z} + \frac{I q}{2\pi c}\ln\left(\frac{\rho_0}{\rho}\right)$$

(*c*)

$$\frac{\partial L}{\partial \phi} = 0 \text{ and } \frac{\partial L}{\partial z} = 0, \text{ so } \phi \text{ and } z \text{ are ignorable coordinates}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\rho}}\right) - \frac{\partial L}{\partial \rho} = 0 \qquad \frac{\partial L}{\partial \rho} = m\rho\dot{\phi}^2 - \frac{q I}{2\pi c}\frac{\dot{z}}{\rho}$$

$$\Longrightarrow \quad m\ddot{\rho} - m\rho\dot{\phi}^2 + \frac{q I}{2\pi c}\frac{\dot{z}}{\rho} = 0$$

(*d*)

$$\frac{\partial L}{\partial \phi} = 0 \Longrightarrow \frac{d}{dt}\frac{\partial L}{\partial \dot{\phi}} = 0 \Longrightarrow \frac{d}{dt}\left(m\rho^2\dot{\phi}\right) = 0 \Longrightarrow m\rho^2\dot{\phi} = c_1$$

where c_1 is a constant determined at $t = 0$

$$\frac{\partial L}{\partial z} = 0 \Longrightarrow \frac{d}{dt}\frac{\partial L}{\partial \dot{z}} = 0 \Longrightarrow \frac{d}{dt}\left[m\dot{z} + \frac{q I}{2\pi c}\ln\left(\frac{\rho_0}{\rho}\right)\right] = 0$$

$$\Longrightarrow m\dot{z} + \frac{q I}{2\pi c}\ln\left(\frac{\rho_0}{\rho}\right) = c_2$$

where c_2 is a constant determined at $t = 0$

(*e*)

$$H = \sum_\alpha \frac{\partial L}{\partial \dot{q}_\alpha}\dot{q}_\alpha - L(q, \dot{q}, t) \quad \text{with } L = \frac{1}{2}m\left(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + \dot{z}^2\right) + \frac{I q \dot{z}}{2\pi c}\ln\left(\frac{\rho_0}{\rho}\right)$$

$$= m\dot{\rho}^2 + m\rho^2\dot{\phi}^2 + m\dot{z}^2 + \frac{I q \dot{z}}{2\pi c}\ln\left(\frac{\rho_0}{\rho}\right) - \left[\frac{1}{2}m\left(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + \dot{z}^2\right) + \frac{I q \dot{z}}{2\pi c}\ln\left(\frac{\rho_0}{\rho}\right)\right]$$

$$= \frac{1}{2} m\left(\dot{\rho}^2 + \rho^2 \dot{\phi}^2 + \dot{z}^2\right) = E_0$$

$\dot{H} = \dfrac{\partial L}{\partial t} = 0$ So H is conserved $H = E_0$ with E_0 determined at $t = 0$

So $\dot{\rho}^2 = \dfrac{2 E_0}{m} - \rho^2 \dot{\phi}^2 - \dot{z}^2$

Noticing that $\begin{cases} m \rho^2 \dot{\phi} = c_1 \\ m \dot{z} + \dfrac{q I}{2 \pi c} \ln\left(\dfrac{\rho_0}{\rho}\right) = c_2 \end{cases} \Longrightarrow \begin{cases} \dot{\phi} = \dfrac{c_1}{m \rho^2} \\ \dot{z} = \dfrac{c_2}{m} - \dfrac{q I}{2 \pi m c} \ln\left(\dfrac{\rho_0}{\rho}\right) \end{cases}$

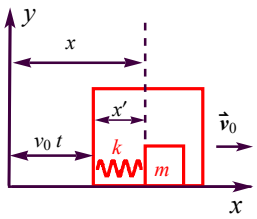
one has

$$\dot{\rho}^2 = \frac{2 E_0}{m} - \rho^2 \dot{\phi}^2 - \dot{z}^2$$
$$= \frac{2 E_0}{m} - \frac{c_1^2}{m^2 \rho^2} - \left[\frac{c_2}{m} + \frac{q I}{2 \pi m c} \ln\left(\frac{\rho}{\rho_0}\right)\right]^2$$

which can be solved to get ρ as a function of time t in principle.

★ 补充题

方盒相对于地面以 v_0 沿 x 向滑行，盒内质点 m 系于原长 l 弹性系数 k 的弹簧并可无摩擦滑动，在地面坐标系 Σ 写出质点的拉氏量 $L(x, \dot{x}, t)$。做广义坐标变换 $x = x(x' t)$，将 L 写成方盒坐标系 Σ′ 坐标 x' 的函数 $L'(x', \dot{x}', t)$。若直接在 Σ′ 系中写出拉格朗日量 L''，问 L' 和 L'' 是否相同？若不同请写出 $L'' - L'$。由 L' 和 L'' 得到的运动方程是否相同？由 L, L' 和 L'' 各自求出其广义能量函数 H, H' 和 H''，三者有何区别？各自的物理意义如何？



Solution:

Σ 系：$T = \dfrac{m \dot{x}^2}{2}$, $V = \dfrac{k}{2}(x' - l)^2 = \dfrac{k}{2}(x - v_0 t - l)^2$,

$$L(x, \dot{x}, t) = T - V = \frac{m \dot{x}^2}{2} - \frac{k}{2}(x - v_0 t - l)^2$$

由 $\begin{cases} x' = x - v_0 t \\ \dot{x}' = \dot{x} - v_0 \end{cases}$ 得 $\begin{cases} x = x' + v_0 t \\ \dot{x} = \dot{x}' + v_0 \end{cases}$ 代入 $L(x, \dot{x}, t)$ 得

$$L' = \frac{m(\dot{x}' + v_0)^2}{2} - \frac{k}{2}(x' - l)^2,$$

直接在 Σ′ 系观察，$L'' = \dfrac{m \dot{x}'^2}{2} - \dfrac{k}{2}(x' - l)^2$,

$$L'' - L' = -\frac{2 m \dot{x}' v_0 + m v_0^2}{2} \neq 0$$

L'' 和 L' 得到相同的运动方程。

因为 $L'' - L' = -\dfrac{2 m \dot{x}' v_0 + m v_0^2}{2} = -\dfrac{d\left(m x' v_0 + \frac{1}{2} m v_0^2 t\right)}{d t} = \dfrac{d F(q, t)}{d t}$

$$F(q, t) = -\left(m x' v_0 + \frac{1}{2} m v_0^2 t\right)$$

由 L 得：$H = \frac{\partial L}{\partial \dot{x}} \dot{x} - L = \frac{m \dot{x}^2}{2} + \frac{k}{2} (x - v_0 t - l)^2$ 实验室坐标系 Σ 的机械能

由 L' 得：$H' = \frac{\partial L'}{\partial \dot{x}'} \dot{x}' - L' = \frac{m \dot{x}'^2}{2} + \frac{k}{2} (x' - l)^2 - \frac{1}{2} m v_0^2$ 与方盒坐标系 Σ′ 中的机械能(见 H'') 差一个常数能量差一个常数无所谓，所以还是 Σ′ 中的机械能

由 L'' 得：$H'' = \frac{\partial L''}{\partial \dot{x}'} \dot{x}' - L' = \frac{m \dot{x}'^2}{2} + \frac{k}{2} (x' - l)^2$ 方盒坐标系 Σ′ 中的机械能